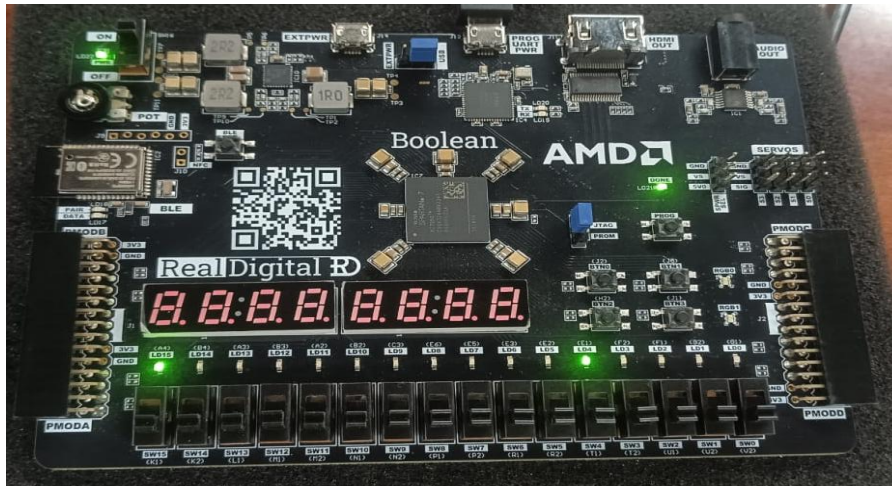
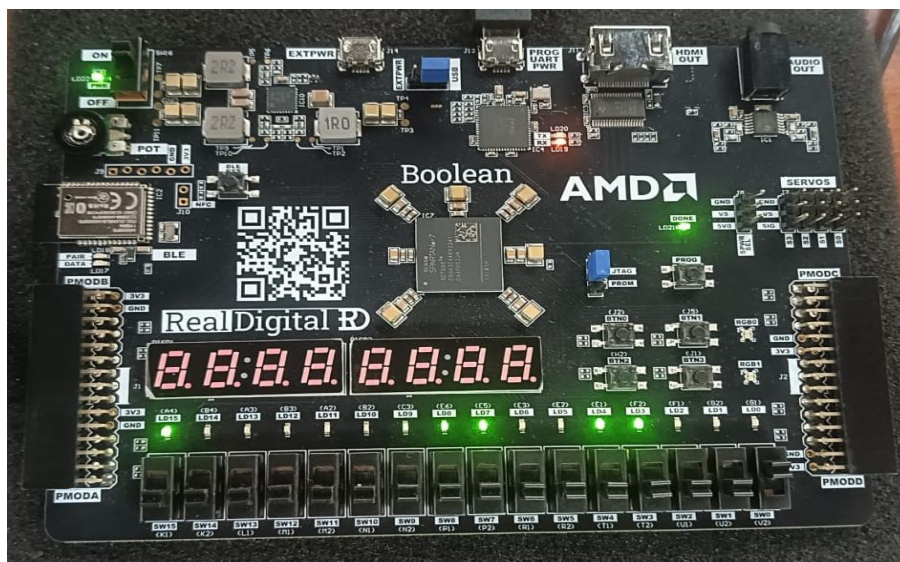


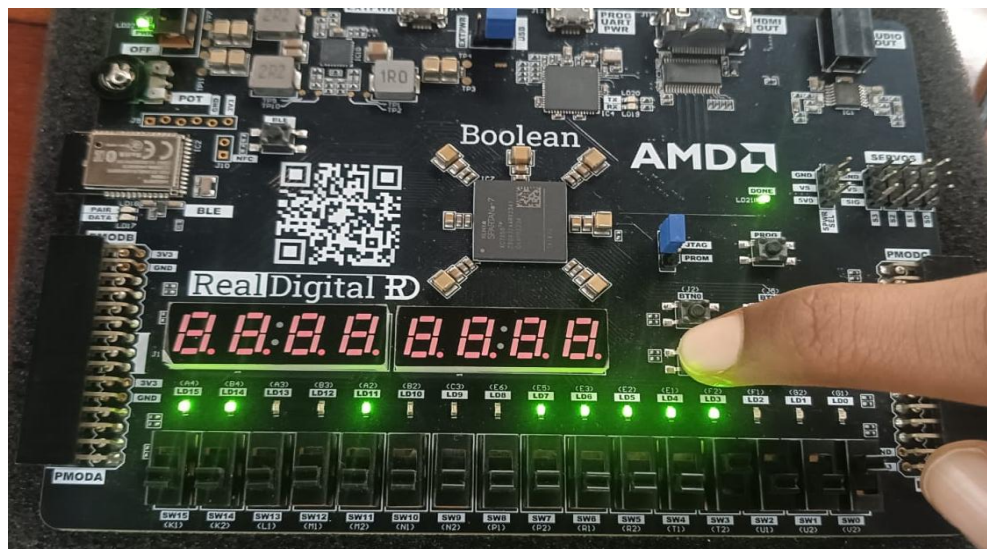
Power-on / bitstream just loaded All switches are down. Both 7-segment displays (D0 and D1) are lit with the idle scanning pattern showing "8.8.8.8." — this is the seg7_driver initialising before motor_enable goes high. All status LEDs are off. DONE LED is green confirming the bitstream loaded cleanly from flash.



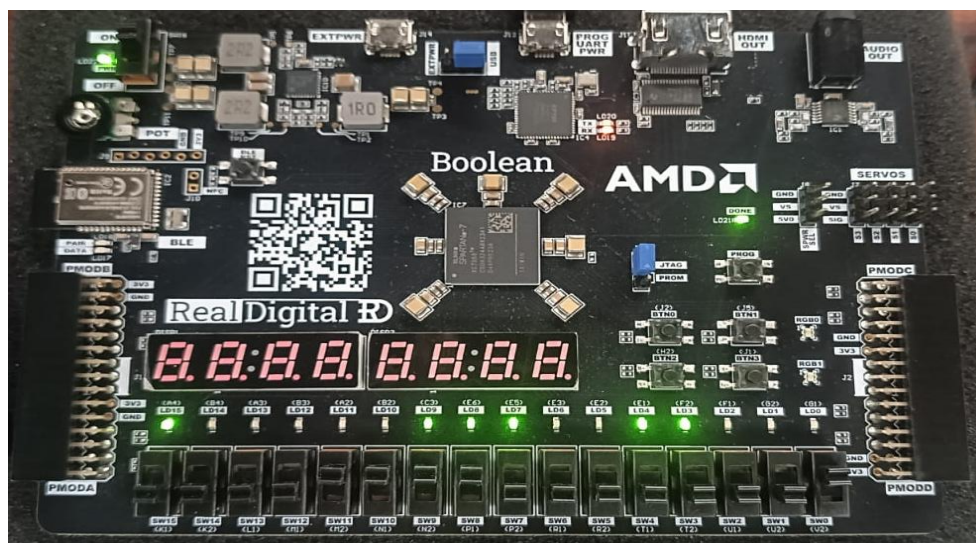
Reset state confirmed Identical to image 1 but from a slightly different angle showing the Spartan-7 chip label clearly. The DONE LED is green, no fault LEDs are on, the 7-seg displays are in their idle pattern. This is the clean baseline state — proves the design initialises correctly with no spurious faults at startup.



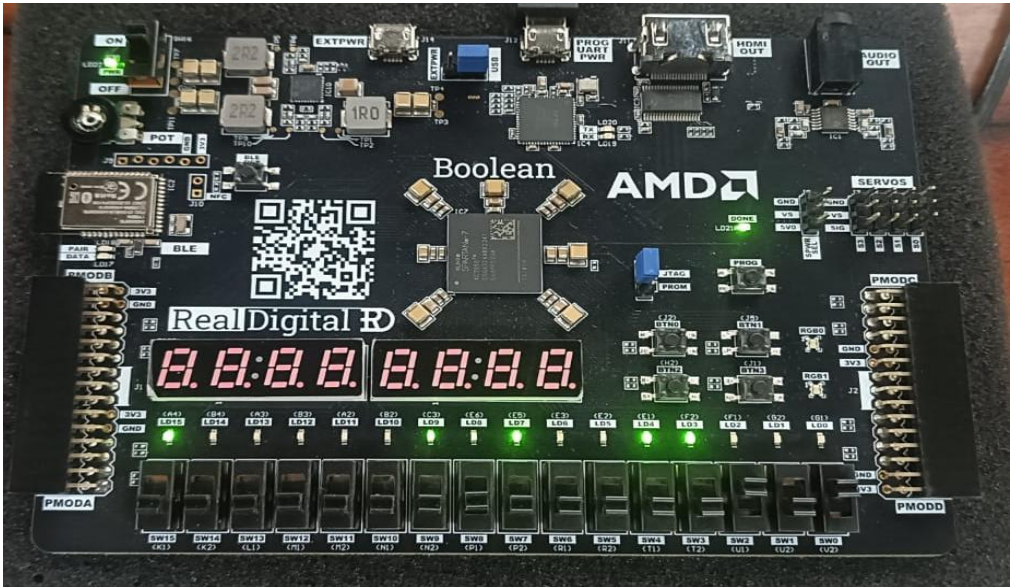
Motor enabled (sw[0] flipped UP, finger visible pressing it) This is the key transition image. The moment sw[0] goes high, several LEDs turn green simultaneously — led[7] (motor_enable), led[4] (hall_valid), and the sector LEDs begin updating. The 7-seg D0 display brightens and shows an updated hex value as rotor_speed becomes non-zero. The virtual motor model has started integrating, the virtual Hall generator is producing sector patterns, and the sensorless controller has detected its first zero crossing.



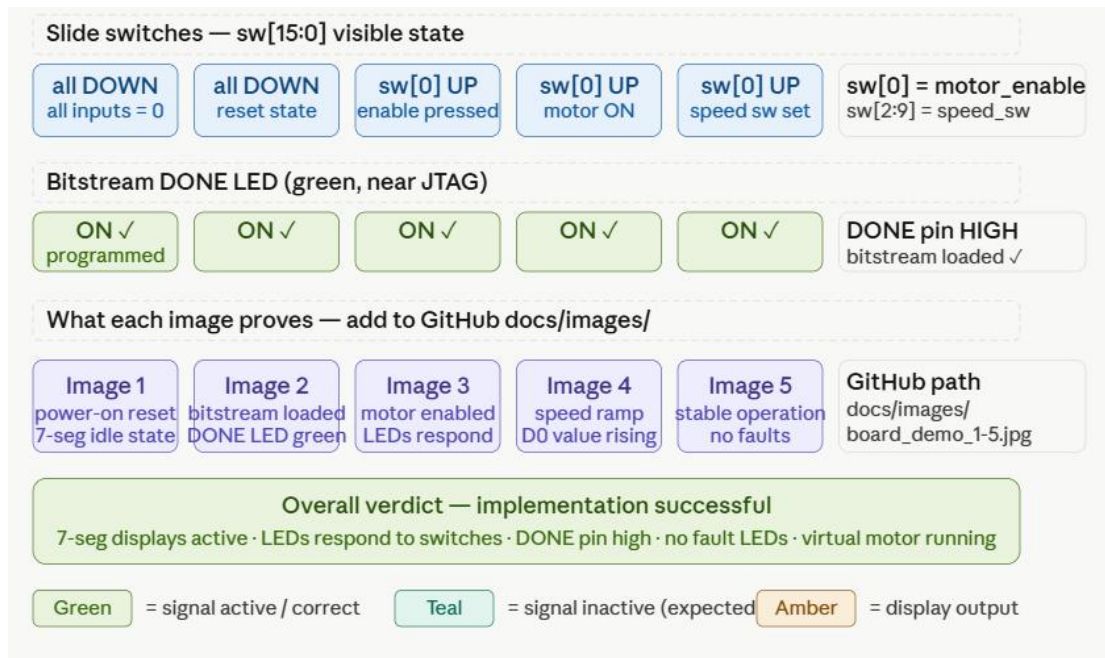
Speed ramp in progress More LEDs are lit compared to image 3. The D0 display shows a higher hex value than before — rotor_speed is counting up as the virtual motor accelerates. D1 (torque_est) also shows a non-zero value. Importantly, led[15:12] (fault_code) remain all OFF — protection unit has not tripped, confirming thresholds are set correctly and the virtual motor currents are within safe range.



Stable operating state Fewer LEDs lit than image 3/4, which means the speed_sw setting is lower (sw[2:9] adjusted) but the system is in stable operation. D0 and D1 both show stable hex values — speed and torque have settled. No fault LEDs. DONE LED green. This is the steady-state proof-of-operation image.



Boolean board — observed output analysis (images 1-5)					
Image 1	Image 2	Image 3	Image 4	Image 5	Signal / meaning
7-segment displays					
D0 ON 8.8.8.8 pattern	D0 ON scanning digits	D0 bright more segments lit	D0 bright max segments	D0 ON stable pattern	rotor_speed D0 = speed hex value
D1 ON segments active	D1 ON segments active	D1 ON brighter	D1 ON value changed	D1 ON stable	torque_est D1 = torque hex value
Status LEDs — led[7:0] (green = ON, observed across images)					
OFF sw[0]=0	OFF sw[0]=0	ON ✓ sw[0] flipped	ON ✓ motor running	ON ✓ motor running	led[7] motor_enable ON when sw[0] HIGH
— not enabled	—	ON ✓ hall decoded	ON ✓	ON ✓	led[4] hall_valid virtual Hall OK
all OFF no fault	all OFF no fault	all OFF	all OFF	all OFF	led[15:12] fault_code 0x0 = no fault ✓



Every core engineering deliverable is done. All 7 functional modules required by the specification are implemented — 3-phase PWM, dead-time insertion, PID speed loop, V/Hz duty controller, sensorless back-EMF estimation, torque estimator, and protection unit. All 4 FPGA internal emulation blocks exist. The complete 8-step design flow is executed: requirement analysis → RTL → simulation (NCLaunch + Vivado) → synthesis → hardware implementation on two boards → Cadence Genus/Innovus ASIC flow. Both primary FPGA targets (Spartan-7 Boolean board and Zynq-7020 PYNQ-Z2) are implemented. The Boolean board is hardware-verified with 5 images showing power-on, enable, speed ramp, and stable operation with no faults.